

NETWORK DESIGN & VERIFICATION REPORT

Multi-Site Enterprise Infrastructure & Campus Architecture

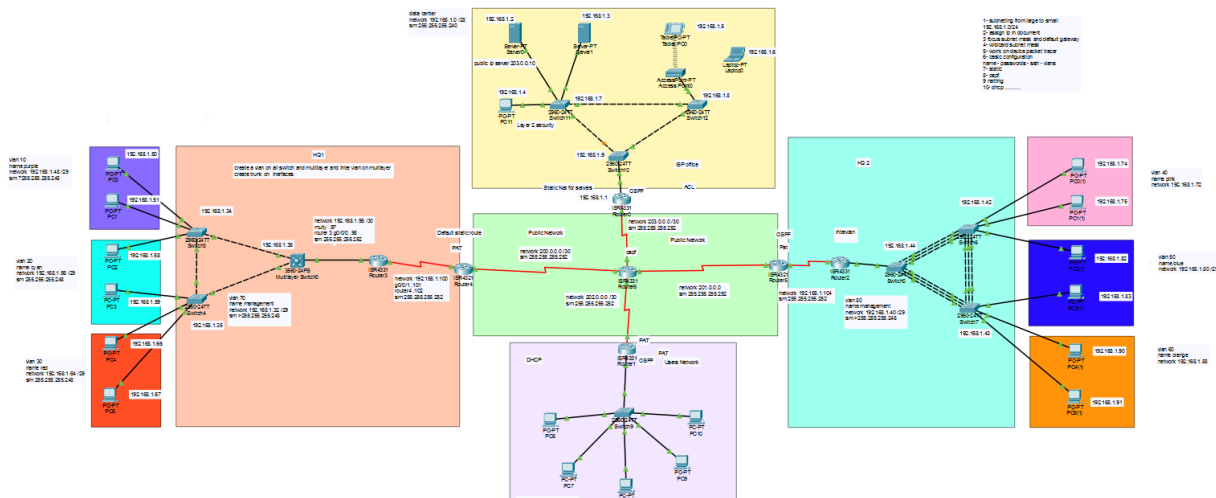


Figure - Complete project topology in Cisco Packet Tracer

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1. Complete IP Addressing & Device Mapping Plan

The master allocation follows VLSM boundaries derived from 192.168.1.0/24, plus the WAN transit networks 200.0.0.0/30 to 203.0.0.0/30. The table below lists every device, interface, address, mask and gateway used in the project.

Device	Interface / SVI	IP Address	Subnet Mask	Gateway	Description
HQ1					
MLS0 (HQ1)	VLAN 10	192.168.1.49	255.255.255.248	N/A	HQ1 Purple VLAN Gateway
MLS0 (HQ1)	VLAN 20	192.168.1.57	255.255.255.248	N/A	HQ1 Cyan VLAN Gateway
MLS0 (HQ1)	VLAN 30	192.168.1.65	255.255.255.248	N/A	HQ1 Red VLAN Gateway
MLS0 (HQ1)	VLAN 70	192.168.1.33	255.255.255.248	N/A	HQ1 Management Gateway
MLS0 (HQ1)	Gig0/1	192.168.1.97	255.255.255.252	192.168.1.98	Uplink to Router3
Router3	Gig0/0	192.168.1.98	255.255.255.252	192.168.1.97	Downlink to MLS0
Router3	Gig0/1	192.168.1.101	255.255.255.252	192.168.1.102	Transit to Router4
Router4	Gig0/0	192.168.1.102	255.255.255.252	192.168.1.101	Link from Router3
Router4	Serial0/0/0	200.0.0.2	255.255.255.252	200.0.0.1	WAN Link to Core Router6
Switch3	VLAN 70	192.168.1.34	255.255.255.248	192.168.1.33	HQ1 Access Switch Management
Switch4	VLAN 70	192.168.1.35	255.255.255.248	192.168.1.33	HQ1 Access Switch Management
PC_Purple_1	NIC	192.168.1.50	255.255.255.248	192.168.1.49	HQ1 User Host (VLAN 10)
PC_Purple_2	NIC	192.168.1.51	255.255.255.248	192.168.1.49	HQ1 User Host (VLAN 10)
PC_Cyan_1	NIC	192.168.1.58	255.255.255.248	192.168.1.57	HQ1 User Host (VLAN 20)
PC_Cyan_2	NIC	192.168.1.59	255.255.255.248	192.168.1.57	HQ1 User Host (VLAN 20)
PC_Red_1	NIC	192.168.1.66	255.255.255.248	192.168.1.65	HQ1 User Host (VLAN 30)
PC_Red_2	NIC	192.168.1.67	255.255.255.248	192.168.1.65	HQ1 User Host (VLAN 30)
HQ2					
Router2	Gig0/0 (sub .40)	192.168.1.73	255.255.255.248	N/A	HQ2 VLAN 40 Gateway (Pink)
Router2	Gig0/0 (sub .50)	192.168.1.81	255.255.255.248	N/A	HQ2 VLAN 50 Gateway (Blue)
Router2	Gig0/0 (sub .60)	192.168.1.89	255.255.255.248	N/A	HQ2 VLAN 60 Gateway (Orange)
Router2	Gig0/0 (sub .80)	192.168.1.41	255.255.255.248	N/A	HQ2 VLAN 80 Management Gateway
Router2	Gig0/1	192.168.1.105	255.255.255.252	192.168.1.106	Transit Link to Router5
Router5	Gig0/0	192.168.1.106	255.255.255.252	192.168.1.105	Link from Router2
Router5	Serial0/0/0	201.0.0.2	255.255.255.252	201.0.0.1	WAN Link to Core Router6
Switch6	VLAN 80	192.168.1.42	255.255.255.248	192.168.1.41	HQ2 Access Switch Management
Switch7	VLAN 80	192.168.1.43	255.255.255.248	192.168.1.41	HQ2 Access Switch Management
PC_Pink_1	NIC	192.168.1.74	255.255.255.248	192.168.1.73	HQ2 User Host (VLAN 40)
PC_Blue_1	NIC	192.168.1.82	255.255.255.248	192.168.1.81	HQ2 User Host (VLAN 50)
PC_Orange_1	NIC	192.168.1.90	255.255.255.248	192.168.1.89	HQ2 User Host (VLAN 60)
SP Office, Users Network and Core					
Router0	Gig0/0	192.168.1.1	255.255.255.240	N/A	SP Office Local Gateway
Router0	Serial0/0/0	203.0.0.2	255.255.255.252	203.0.0.1	Public WAN Link to Core Router6
Server0	NIC	192.168.1.2	255.255.255.240	192.168.1.1	Internal Server (NAT mapped)
Server1	NIC	192.168.1.3	255.255.255.240	192.168.1.1	Web/DNS Internal Server
Router1	Gig0/0	192.168.1.17	255.255.255.240	N/A	Users Network Gateway
Router1	Serial0/0/0	202.0.0.2	255.255.255.252	202.0.0.1	WAN Link to Core Router6
Core Router6	Serial0/0/0	200.0.0.1	255.255.255.252	200.0.0.2	Core Link to HQ1 (Router4)
Core Router6	Serial0/0/1	201.0.0.1	255.255.255.252	201.0.0.2	Core Link to HQ2 (Router5)
Core Router6	Serial0/1/0	202.0.0.1	255.255.255.252	202.0.0.2	Core Link to Users (Router1)
Core Router6	Serial0/1/1	203.0.0.1	255.255.255.252	203.0.0.2	Core Link to SP Office (Router0)

2. HQ1 - Topology & Design

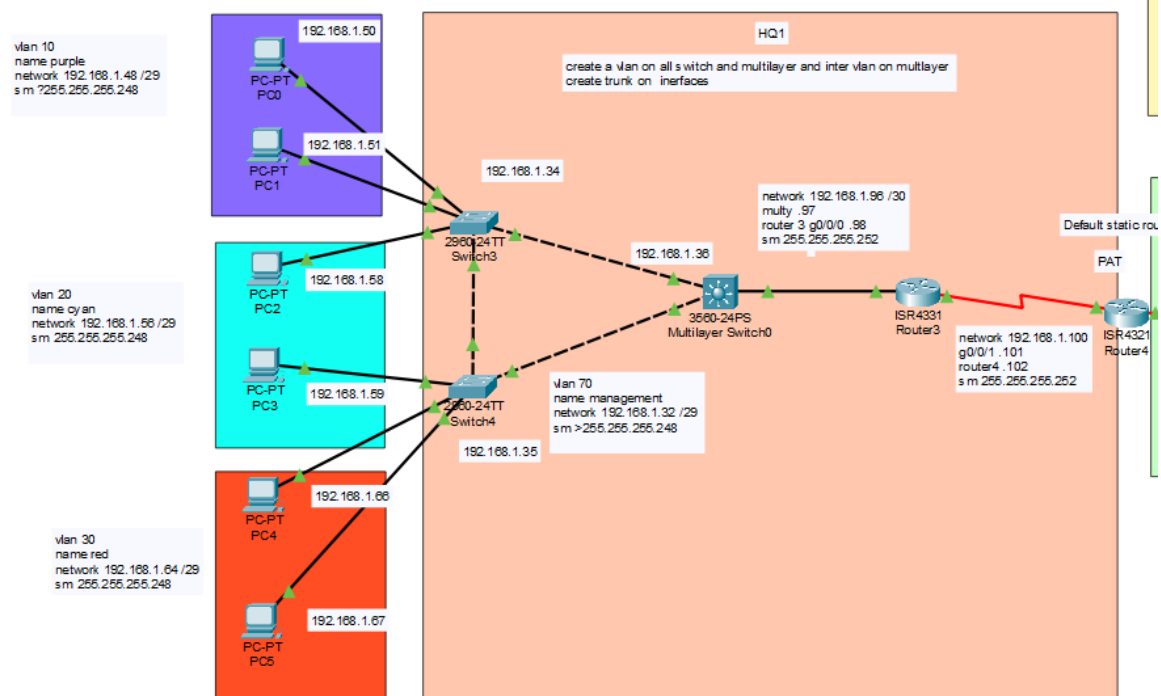


Figure 1 - HQ1 topology (left site)

Overview

HQ1 is the left-hand campus. Three departmental VLANs (Purple, Cyan, Red) and one management VLAN are carried from two access switches (Switch3, Switch4) over trunk links to a Multilayer Switch (MLS0, Cisco 3560-24PS). MLS0 performs inter-VLAN routing with one SVI per VLAN, then hands traffic to Router3 over a routed /30 link. Router3 connects to Router4, the site's WAN edge router, which reaches the core network through 200.0.0.0/30.

VLAN plan

VLAN	Name	Network	Gateway	Hosts / Devices
10	Purple	192.168.1.48/29	192.168.1.49	PC0 - 192.168.1.50, PC1 - 192.168.1.51
20	Cyan	192.168.1.56/29	192.168.1.57	PC2 - 192.168.1.58, PC3 - 192.168.1.59
30	Red	192.168.1.64/29	192.168.1.65	PC4 - 192.168.1.66, PC5 - 192.168.1.67
70	Management	192.168.1.32/29	192.168.1.33	Switch3 - 192.168.1.34, Switch4 - 192.168.1.35

What was implemented

- VLANs 10, 20, 30 and 70 created on both access switches and on the multilayer switch, with trunk links between all switches.
- Inter-VLAN routing on MLS0 (**ip routing** plus one SVI per VLAN) so every host uses its VLAN SVI as default gateway.
- Routed uplink: MLS0 Gig0/1 (192.168.1.97) to Router3 Gig0/0 (192.168.1.98) on 192.168.1.96/30.
- Transit link: Router3 Gig0/1 (192.168.1.101) to Router4 Gig0/0 (192.168.1.102) on 192.168.1.100/30.
- Router4 is the WAN edge: serial link 200.0.0.2/30 towards Core Router6, with the default static route and PAT noted in the topology.
- OSPFv2 (process 1, area 0) on MLS0, Router3 and Router4 to advertise all HQ1 networks.

2.1 HQ1 - CLI Configuration

MLS0 - Multi-Layer Switch (HQ1 Core)

```
enable
configure terminal
hostname MLS0
vlan 10
  name Purple_VLAN
vlan 20
  name Cyan_VLAN
vlan 30
  name Red_VLAN
vlan 70
  name Management_VLAN
exit
ip routing
interface vlan 10
  ip address 192.168.1.49 255.255.255.248
  no shutdown
interface vlan 20
  ip address 192.168.1.57 255.255.255.248
  no shutdown
interface vlan 30
  ip address 192.168.1.65 255.255.255.248
  no shutdown
interface vlan 70
  ip address 192.168.1.33 255.255.255.248
  no shutdown
interface GigabitEthernet0/1
  no switchport
  ip address 192.168.1.97 255.255.255.252
  no shutdown
interface FastEthernet0/1
  switchport mode trunk
  switchport trunk allowed vlan 10,20,30,70
interface FastEthernet0/2
  switchport mode trunk
  switchport trunk allowed vlan 10,20,30,70
router ospf 1
  router-id 1.1.1.1
  network 192.168.1.48 0.0.0.7 area 0
  network 192.168.1.56 0.0.0.7 area 0
  network 192.168.1.64 0.0.0.7 area 0
  network 192.168.1.32 0.0.0.7 area 0
  network 192.168.1.96 0.0.0.3 area 0
exit
do write memory
```

Router3 - HQ1 Intermediate Router

```
enable
configure terminal
hostname Router3
interface GigabitEthernet0/0
  ip address 192.168.1.98 255.255.255.252
  no shutdown
interface GigabitEthernet0/1
  ip address 192.168.1.101 255.255.255.252
  no shutdown
router ospf 1
  router-id 3.3.3.3
  network 192.168.1.96 0.0.0.3 area 0
  network 192.168.1.100 0.0.0.3 area 0
exit
do write memory
```

Router4 - HQ1 WAN Edge Router

```
enable
configure terminal
hostname Router4
interface GigabitEthernet0/0
 ip address 192.168.1.102 255.255.255.252
 no shutdown
interface Serial0/0/0
 ip address 200.0.0.2 255.255.255.252
 no shutdown
router ospf 1
 router-id 4.4.4.4
 network 192.168.1.100 0.0.0.3 area 0
 network 200.0.0.0 0.0.0.3 area 0
exit
do write memory
```

3. HQ2 - Topology & Design

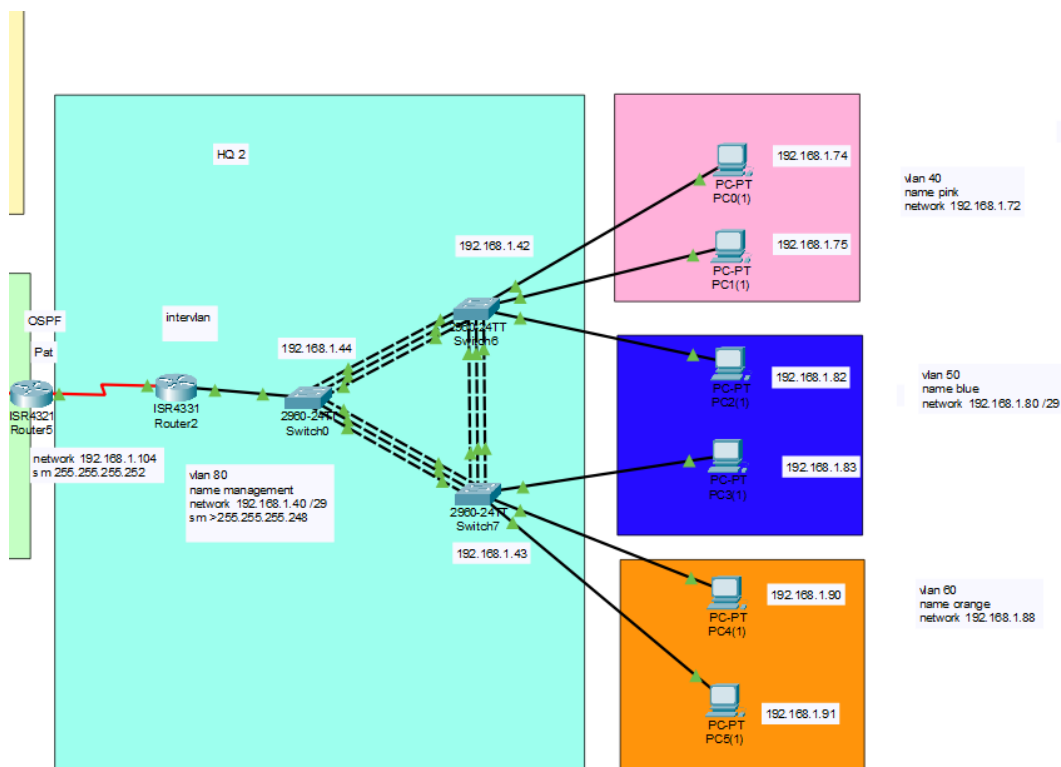


Figure 2 - HQ2 topology (right site)

Overview

HQ2 is the right-hand campus. It uses the router-on-a-stick model: Router2 has a single physical LAN interface (Gig0/0) split into dot1Q sub-interfaces, one per VLAN. Access switches Switch6 and Switch7 are connected to each other and to the distribution switch (Switch0) with multiple parallel trunk links, and Switch0 connects to Router2. Router2 connects to Router5, the WAN edge router, which reaches the core over 201.0.0.0/30.

VLAN plan

VLAN	Name	Network	Gateway	Hosts / Devices
40	Pink	192.168.1.72/29	192.168.1.73	PC0(1) - 192.168.1.74, PC1(1) - 192.168.1.75
50	Blue	192.168.1.80/29	192.168.1.81	PC2(1) - 192.168.1.82, PC3(1) - 192.168.1.83
60	Orange	192.168.1.88/29	192.168.1.89	PC4(1) - 192.168.1.90, PC5(1) - 192.168.1.91
80	Management	192.168.1.40/29	192.168.1.41	Switch6 - 192.168.1.42, Switch7 - 192.168.1.43

What was implemented

- VLANs 40, 50, 60 and 80 created on the HQ2 switches, with trunk links between switches and towards Router2.
- Inter-VLAN routing with router-on-a-stick: sub-interfaces Gig0/0.40, .50, .60 and .80 with dot1Q encapsulation.
- Management VLAN 80: each access switch has a management address (Switch6 - .42, Switch7 - .43) with gateway 192.168.1.41. Remote SSH access to Switch6 was verified (see the verification section).
- Transit link: Router2 Gig0/1 (192.168.1.105) to Router5 Gig0/0 (192.168.1.106) on 192.168.1.104/30.
- Router5 is the WAN edge: serial link 201.0.0.2/30 towards Core Router6, with PAT noted in the topology.
- OSPFv2 (process 1, area 0) on Router2 and Router5 to advertise all HQ2 networks.

3.1 HQ2 - CLI Configuration

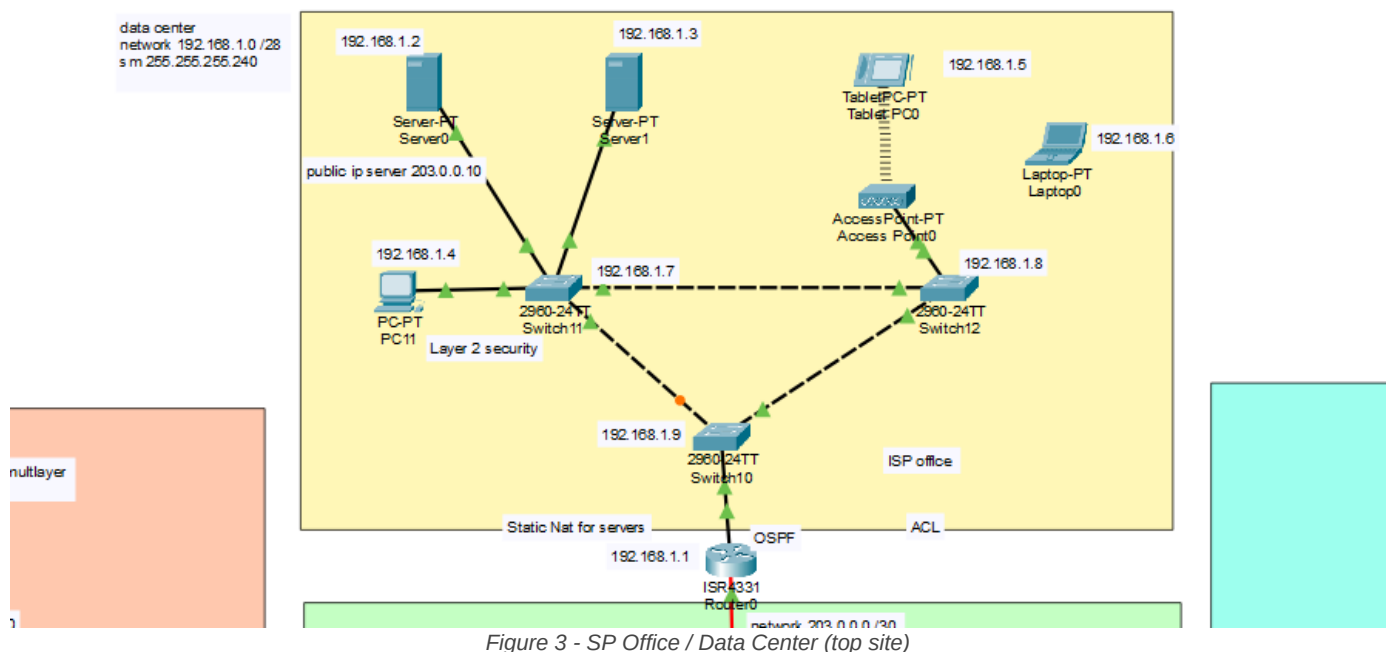
Router2 - HQ2 Router-on-a-Stick Gateway

```
enable
configure terminal
hostname Router2
interface GigabitEthernet0/0
no shutdown
interface GigabitEthernet0/0.40
encapsulation dot1Q 40
ip address 192.168.1.73 255.255.255.248
interface GigabitEthernet0/0.50
encapsulation dot1Q 50
ip address 192.168.1.81 255.255.255.248
interface GigabitEthernet0/0.60
encapsulation dot1Q 60
ip address 192.168.1.89 255.255.255.248
interface GigabitEthernet0/0.80
encapsulation dot1Q 80
ip address 192.168.1.41 255.255.255.248
interface GigabitEthernet0/1
ip address 192.168.1.105 255.255.255.252
no shutdown
router ospf 1
router-id 2.2.2.2
network 192.168.1.72 0.0.0.7 area 0
network 192.168.1.80 0.0.0.7 area 0
network 192.168.1.88 0.0.0.7 area 0
network 192.168.1.40 0.0.0.7 area 0
network 192.168.1.104 0.0.0.3 area 0
exit
do write memory
```

Router5 - HQ2 WAN Edge Router

```
enable
configure terminal
hostname Router5
interface GigabitEthernet0/0
ip address 192.168.1.106 255.255.255.252
no shutdown
interface Serial0/0/0
ip address 201.0.0.2 255.255.255.252
no shutdown
router ospf 1
router-id 5.5.5.5
network 192.168.1.104 0.0.0.3 area 0
network 201.0.0.0 0.0.0.3 area 0
exit
do write memory
```

4. SP Office (Data Center) - Topology & Design



Overview

The top site hosts the servers and the service-provider office equipment on the 192.168.1.0/28 data-center network. Three switches (Switch10, Switch11, Switch12) are interconnected with trunk links and connect to Router0, which is the gateway of the site and its link to the public network (203.0.0.0/30). Server0 is published to the outside world through static NAT using the public address 203.0.0.10.

Addressing

Device	Address	Notes
Router0 Gig0/0	192.168.1.1/28	Site gateway (NAT inside)
Server0	192.168.1.2	Internal server - public address 203.0.0.10 (static NAT)
Server1	192.168.1.3	Web/DNS internal server
Router0 Serial0/0/0	203.0.0.2/30	Link to Core Router6 (203.0.0.1) - NAT outside

What was implemented

- Data-center network 192.168.1.0/28 with Router0 as default gateway (192.168.1.1).
- Static NAT on Router0: **ip nat inside source static 192.168.1.2 203.0.0.10**, so Server0 can be reached from other sites using the public address.
- Router0 Gig0/0 marked as NAT inside and Serial0/0/0 as NAT outside.
- The topology also shows Layer 2 security on the switches and an ACL on the site edge, in addition to the wireless access point serving the tablet.
- OSPFv2 (process 1, area 0) on Router0 to advertise the data-center network and the 203.0.0.0/30 link.

4.1 SP Office - CLI Configuration

Router0 - SP Office and Static NAT Gateway

```
enable
configure terminal
hostname Router0
interface GigabitEthernet0/0
 ip address 192.168.1.1 255.255.255.240
 ip nat inside
 no shutdown
interface Serial0/0/0
 ip address 203.0.0.2 255.255.255.252
 ip nat outside
 no shutdown
! Static NAT Translation for Internal Server0
ip nat inside source static 192.168.1.2 203.0.0.10
router ospf 1
 router-id 10.10.10.10
 network 192.168.1.0 0.0.0.15 area 0
 network 203.0.0.0 0.0.0.3 area 0
exit
do write memory
```

NAT verification (expected output)

Executed on Router0: show ip nat translations

Pro	Inside global	Inside local	Outside local	Outside global
---	203.0.0.10	192.168.1.2	---	---

5. Users Network - Topology & Design

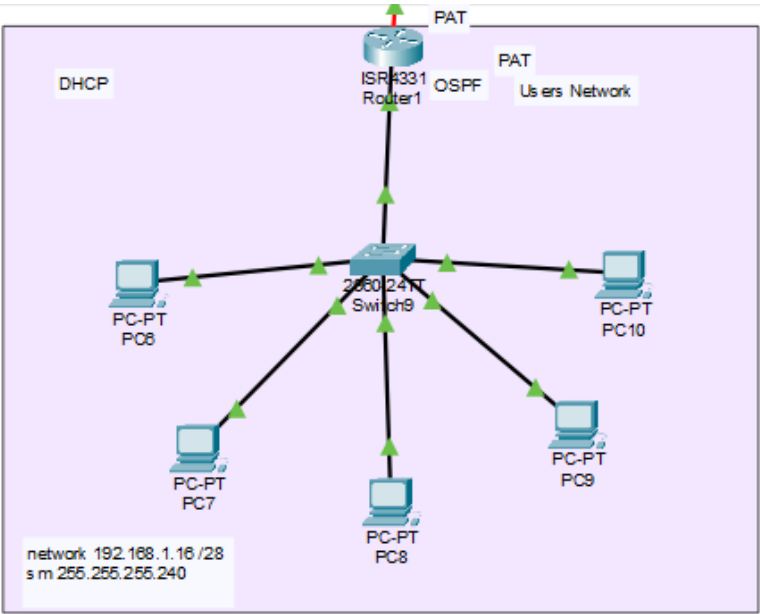


Figure 4 - Users Network (bottom site)

Overview

The bottom site is the Users Network: five PCs (PC6 to PC10) connected to Switch9, which uplinks to Router1. Router1 is both the default gateway and the DHCP server of the site, and it connects to the core through 202.0.0.0/30.

Addressing

Item	Value
Network	192.168.1.16/28 (255.255.255.240)
Default gateway	Router1 Gig0/0 - 192.168.1.17
DHCP pool	USERS_POOL - default-router 192.168.1.17, DNS server 192.168.1.3
Excluded address	192.168.1.17 (gateway)
WAN link	Router1 Serial0/0/0 - 202.0.0.2/30 to Core Router6 (202.0.0.1)

What was implemented

- DHCP service on Router1, so the PCs receive address, gateway and DNS automatically.
- The gateway address is excluded from the pool to avoid conflicts.
- PAT is noted on Router1 in the topology for internet-bound traffic.
- OSPFv2 (process 1, area 0) on Router1 to advertise 192.168.1.16/28 and 202.0.0.0/30.

5.1 Users Network - CLI Configuration

Router1 - Users Network Gateway and DHCP Server

```
enable
configure terminal
hostname Router1
interface GigabitEthernet0/0
  ip address 192.168.1.17 255.255.255.240
  no shutdown
interface Serial0/0/0
  ip address 202.0.0.2 255.255.255.252
  no shutdown
! DHCP Configuration for Dynamic Users Pool
ip dhcp excluded-address 192.168.1.17
ip dhcp pool USERS_POOL
  network 192.168.1.16 255.255.255.240
  default-router 192.168.1.17
  dns-server 192.168.1.3
exit
router ospf 1
  router-id 11.11.11.11
  network 192.168.1.16 0.0.0.15 area 0
  network 202.0.0.0 0.0.0.3 area 0
exit
do write memory
```

6. Public Network (Core) - Topology & Design

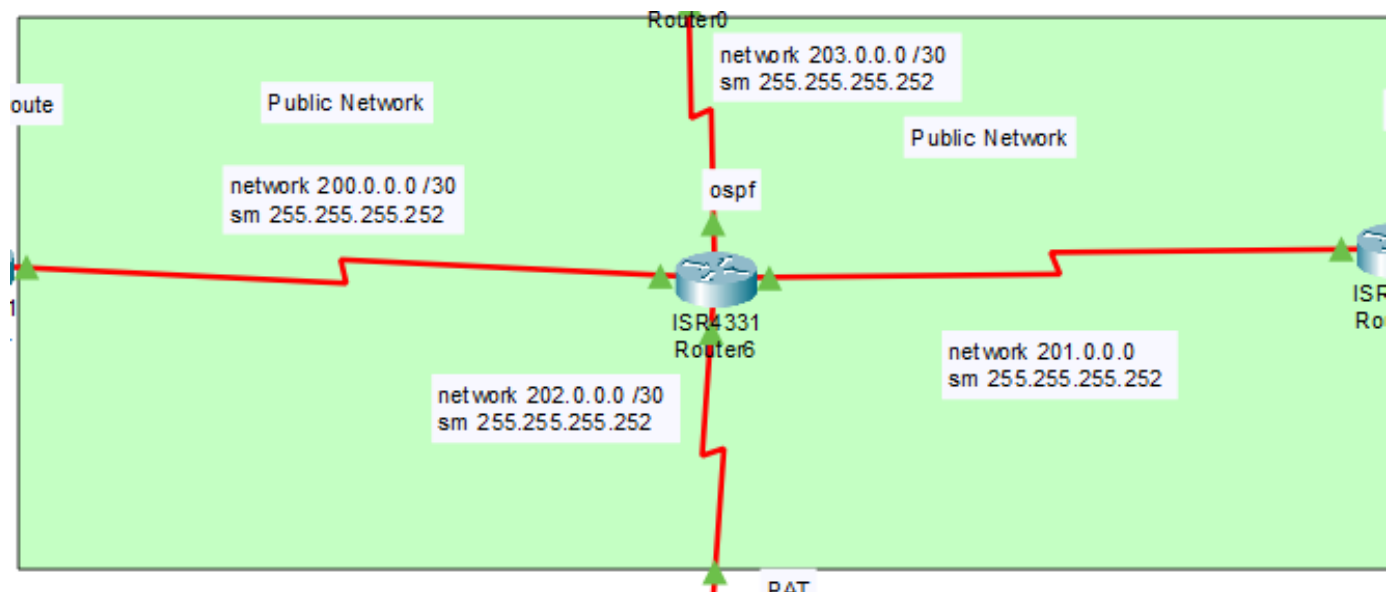


Figure 5 - Public network with Core Router6 (center)

Overview

The green area in the center simulates the public network / internet. Core Router6 has four serial links, one to each site edge router, and runs OSPF so that all sites learn each other's routes through it. Router6 is the DCE side of every link, so each serial interface has a clock rate.

Public links

Network	Core Router6 side	Remote side	Connected site
200.0.0.0/30	Serial0/0/0 - 200.0.0.1	Router4 - 200.0.0.2	HQ1
201.0.0.0/30	Serial0/0/1 - 201.0.0.1	Router5 - 201.0.0.2	HQ2
202.0.0.0/30	Serial0/1/0 - 202.0.0.1	Router1 - 202.0.0.2	Users Network
203.0.0.0/30	Serial0/1/1 - 203.0.0.1	Router0 - 203.0.0.2	SP Office (Data Center)

What was implemented

- Four /30 point-to-point serial links with **clock rate 64000** on the DCE side.
- OSPFv2 (process 1, area 0) advertising all four public /30 networks.
- Static NAT (Router0) and PAT (site edges) are what allow private 192.168.1.0/24 hosts to be reached or to reach the public side.

6.1 Core Router6 - CLI Configuration

Core_Router6 - Central OSPF Transit Core

```
enable
configure terminal
hostname Core_Router6
interface Serial0/0/0
 ip address 200.0.0.1 255.255.255.252
 clock rate 64000
 no shutdown
interface Serial0/0/1
 ip address 201.0.0.1 255.255.255.252
 clock rate 64000
 no shutdown
interface Serial0/1/0
 ip address 202.0.0.1 255.255.255.252
 clock rate 64000
 no shutdown
interface Serial0/1/1
 ip address 203.0.0.1 255.255.255.252
 clock rate 64000
 no shutdown
router ospf 1
 router-id 6.6.6.6
 network 200.0.0.0 0.0.0.3 area 0
 network 201.0.0.0 0.0.0.3 area 0
 network 202.0.0.0 0.0.0.3 area 0
 network 203.0.0.0 0.0.0.3 area 0
exit
do write memory
```

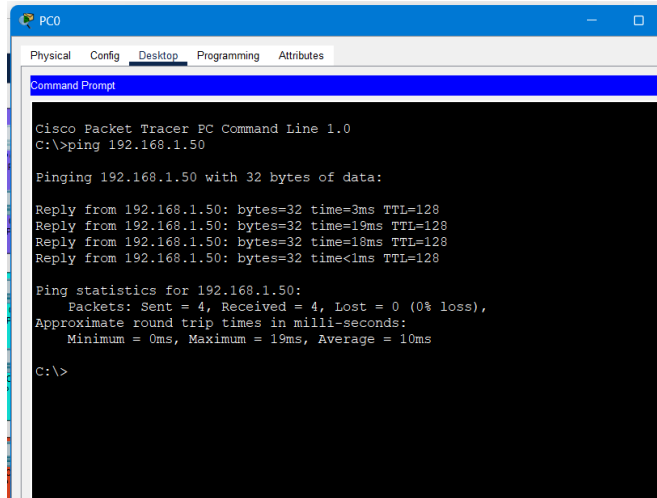
OSPF neighbor validation (expected output)

Executed on Core_Router6: show ip ospf neighbor. Expected: four neighbors (4.4.4.4, 5.5.5.5, 11.11.11.11, 10.10.10.10) in FULL state.

7. Verification & Testing

The following tests were executed in Cisco Packet Tracer to confirm that VLANs, inter-VLAN routing, OSPF routing, static NAT and remote management work as designed.

Test 1 - Host configuration check (HQ1)



```

Cisco Packet Tracer PC Command Line 1.0
C:\>ping 192.168.1.50

Pinging 192.168.1.50 with 32 bytes of data:

Reply from 192.168.1.50: bytes=32 time=3ms TTL=128
Reply from 192.168.1.50: bytes=32 time=19ms TTL=128
Reply from 192.168.1.50: bytes=32 time=18ms TTL=128
Reply from 192.168.1.50: bytes=32 time<1ms TTL=128

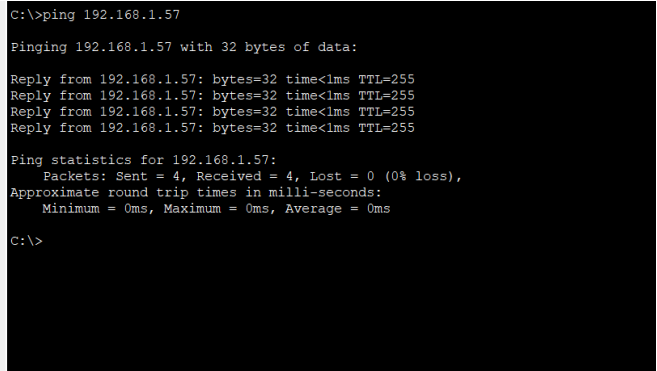
Ping statistics for 192.168.1.50:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 19ms, Average = 10ms

C:\>

```

PC0 (VLAN 10) pings its own address 192.168.1.50: 4 sent, 4 received, 0% loss. Confirms the host IP configuration is active.

Test 2 - Reach the VLAN gateway (HQ1)



```

C:\>ping 192.168.1.57

Pinging 192.168.1.57 with 32 bytes of data:

Reply from 192.168.1.57: bytes=32 time<1ms TTL=255
Reply from 192.168.1.57: bytes=32 time<1ms TTL=255
Reply from 192.168.1.57: bytes=32 time<1ms TTL=255
Reply from 192.168.1.57: bytes=32 time<1ms TTL=255

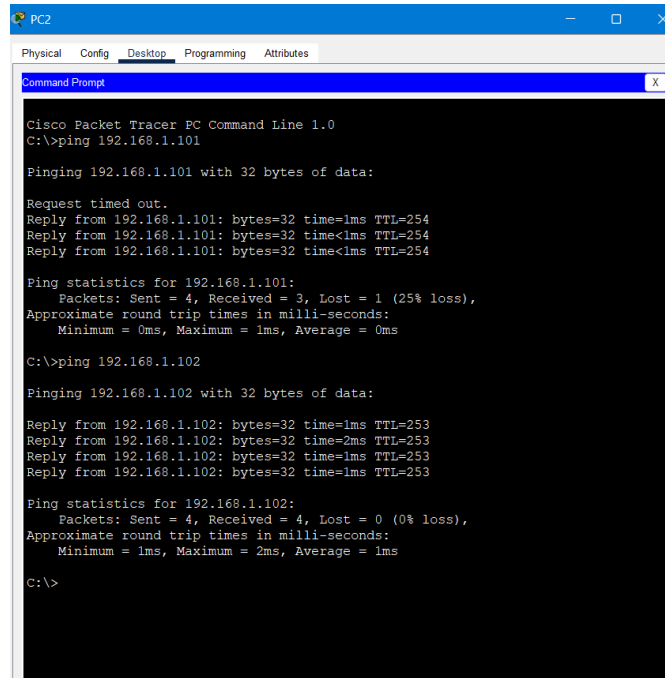
Ping statistics for 192.168.1.57:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\>

```

Ping to 192.168.1.57, the VLAN 20 SVI on MLS0: 4/4 replies, 0% loss, TTL 255 (a network device). Confirms inter-VLAN gateway reachability.

Test 3 - Reach the HQ1 routers (Router3 and Router4)



```

Cisco Packet Tracer PC Command Line 1.0
C:\>ping 192.168.1.101

Pinging 192.168.1.101 with 32 bytes of data:

Request timed out.
Reply from 192.168.1.101: bytes=32 time=1ms TTL=254
Reply from 192.168.1.101: bytes=32 time<1ms TTL=254
Reply from 192.168.1.101: bytes=32 time<1ms TTL=254

Ping statistics for 192.168.1.101:
    Packets: Sent = 4, Received = 3, Lost = 1 (25% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 1ms, Average = 0ms

C:\>ping 192.168.1.102

Pinging 192.168.1.102 with 32 bytes of data:

Reply from 192.168.1.102: bytes=32 time=1ms TTL=253
Reply from 192.168.1.102: bytes=32 time=2ms TTL=253
Reply from 192.168.1.102: bytes=32 time=1ms TTL=253
Reply from 192.168.1.102: bytes=32 time=1ms TTL=253

Ping statistics for 192.168.1.102:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 1ms, Maximum = 2ms, Average = 1ms

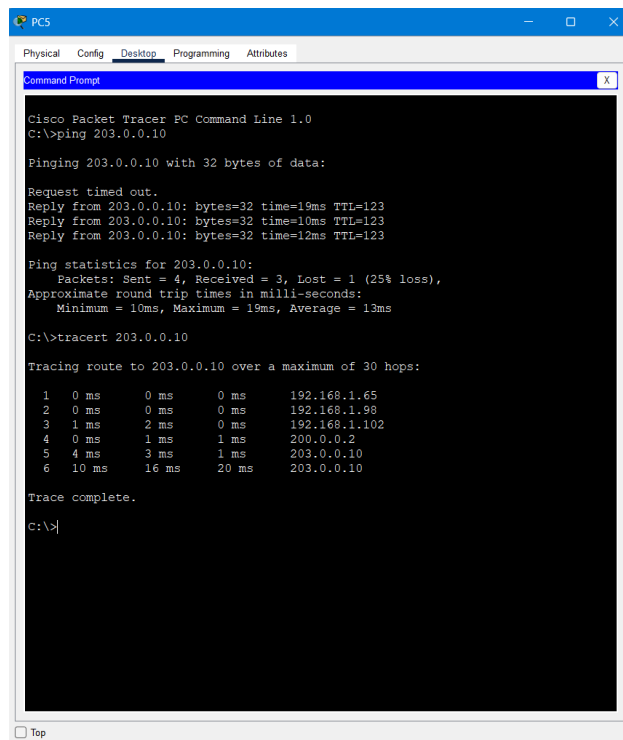
C:\>

```

PC2 pings 192.168.1.101 (Router3) and 192.168.1.102 (Router4). Router4 replies 4/4 with 0% loss; Router3 replies 3/4 because the first request timed out while ARP resolution completed. Confirms the chain PC to MLS0 to Router3 to Router4.

7.1 Public IP Access & Traceroute

Test 4 - Access the server through its public IP (203.0.0.10)



```

Cisco Packet Tracer PC Command Line 1.0
C:\>ping 203.0.0.10

Pinging 203.0.0.10 with 32 bytes of data:

Request timed out.
Reply from 203.0.0.10: bytes=32 time=19ms TTL=123
Reply from 203.0.0.10: bytes=32 time=10ms TTL=123
Reply from 203.0.0.10: bytes=32 time=12ms TTL=123

Ping statistics for 203.0.0.10:
    Packets: Sent = 4, Received = 3, Lost = 1 (25% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 10ms, Maximum = 19ms, Average = 13ms

C:\>tracert 203.0.0.10

Tracing route to 203.0.0.10 over a maximum of 30 hops:

  0  0 ms    0 ms    0 ms    192.168.1.65
  1  0 ms    0 ms    0 ms    192.168.1.98
  2  1 ms    2 ms    0 ms    192.168.1.102
  3  0 ms    1 ms    1 ms    200.0.0.2
  4  4 ms    3 ms    1 ms    203.0.0.10
  5  10 ms   16 ms   20 ms   203.0.0.10

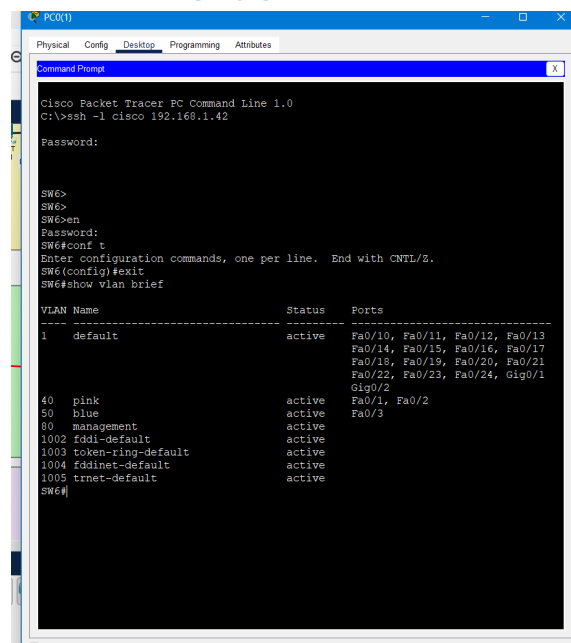
Trace complete.

C:\>

```

PC5 (VLAN 30, HQ1) pings 203.0.0.10: 3 of 4 replies (first request timed out), average 13 ms. The traceroute shows the path: 192.168.1.65 (MLS0 VLAN 30 SVI), 192.168.1.98 (Router3), 192.168.1.102 (Router4), then the 200.0.0.0/30 WAN segment towards the core, and finally 203.0.0.10. This confirms OSPF routing across the core and the static NAT of Server0.

Test 5 - Remote management over SSH (HQ2)



```

Cisco Packet Tracer PC Command Line 1.0
C:\>ssh -l cisco 192.168.1.42

Password:

SW6>
SW6>
SW6>en
SW6#conf t
Enter configuration commands, one per line. End with CNTL/Z.
SW6(config)#exit
SW6#show vlan brief

VLAN Name                Status    Ports
-----
1    default                active    Fa0/10, Fa0/11, Fa0/12, Fa0/13,
                                           Fa0/14, Fa0/15, Fa0/16, Fa0/17,
                                           Fa0/18, Fa0/19, Fa0/20, Fa0/21,
                                           Fa0/22, Fa0/23, Fa0/24, Gig0/1
                                           Gig0/2
40   pink                    active    Fa0/1, Fa0/2
50   blue                    active    Fa0/3
80   Management              active
1002 fddi-default            active
1003 token-ring-default    active
1004 fddinet-default        active
1005 trnet-default          active
SW6#

```

PC0(1) in HQ2 (VLAN 40) opens an SSH session to Switch6 at its management address 192.168.1.42, enters privileged mode and runs show vlan brief: VLAN 40 (pink) on Fa0/1-2, VLAN 50 (blue) on Fa0/3 and VLAN 80 (management) are active.

8. Test Summary & Troubleshooting Log

#	Test	Source / Destination	Result
1	Local host configuration	PC0 to 192.168.1.50	Success - 0% loss
2	VLAN gateway	HQ1 host to 192.168.1.57 (MLS0 SVI VLAN 20)	Success - 0% loss
3	HQ1 routers	PC2 to 192.168.1.101 / 192.168.1.102	Success - 25% / 0% loss (first packet, ARP)
4	Public NAT address + traceroute	PC5 to 203.0.0.10	Success - 25% loss (first packet), path traced
5	SSH management	PC0(1) to Switch6 192.168.1.42	Success - login and show vlan brief

Issue resolution log

Issue	Symptom	Root cause	Fix applied
1. VLAN communication failure (HQ1)	Hosts in VLAN 10 could not reach hosts in VLAN 20.	IP routing was disabled on MLS0.	Enabled ip routing globally on MLS0 and brought up the SVIs for VLAN 10 and VLAN 20.
2. Remote access to Server0 via WAN	Packets from outside the SP Office targeting 192.168.1.2 were dropped at Core Router6.	A private, non-routable IP address was used across the public WAN.	Configured static NAT 192.168.1.2 to 203.0.0.10 on Router0 and pointed remote access to 203.0.0.10.
3. Dynamic hosts missing default gateway	Users Network hosts received DHCP addresses but could not reach external networks.	The default-router parameter was missing from USERS_POOL on Router1.	Added default-router 192.168.1.17 to the USERS_POOL configuration.

Conclusion

The multi-site network was designed with VLSM addressing, VLAN segmentation, inter-VLAN routing (Layer 3 switch at HQ1 and router-on-a-stick at HQ2), single-area OSPF across all sites and the core, static NAT for the published server, DHCP for the Users Network and SSH-based management. The verification tests confirm that every segment is reachable and that the server is accessible through its public address.